

# AKASHDEEP SINGH JIDA

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## EDUCATION

### Bachelor of Science, Electrical Engineering

GPA: 3.25/4.00

California State University, Sacramento, CA

Graduated May 2020

### Associates in Mathematics

Fresno City College, Fresno, CA

Aug 2015 – Dec 2017

## TECHNICAL SKILLS

- |                |                    |                 |                      |
|----------------|--------------------|-----------------|----------------------|
| • Python       | • Cadence Virtuoso | • AutoCAD       | • Function Generator |
| • Verilog/VHDL | • Xilinx           | • Quartus       | • LaTeX              |
| • C/C++        | • ModelSim         | • Visual Studio | • FPGA               |
| • Matlab       | • PSpice           | • PCB designing | • Raspberry Pi       |
| • Arduino UNO  | • STM32            | • Oscilloscope  |                      |

## RELEVANT COURSEWORK

- |                         |                            |
|-------------------------|----------------------------|
| • Advanced Logic Design | • Computer Hardware Design |
| • CMOS and VLSI         | • Power Control Drivers    |
| • Signals and Systems   | • Applied Electromagnetics |
| • Network Analysis      | • Electronics              |

## WORK EXPERIENCE

### California State University, Sacramento

#### Research Assistant

June 2020 – Present

- Assistant in literature review, technical writing and research on safety of mmWave communication.
- Ongoing work includes research on safeguarding 5G communication and publishing research findings.

### Fresno City College, Fresno

#### Instructional Student Assistant – Tutor

Aug 2017 – Dec 2017

- Provided appropriate tutorials to meet each student's need.
- Marked and provided feedbacks to the students.

## PROJECTS

### Hydroponics System (*Senior Design Project*)

Aug 2020 - May 2020

- Designed, developed, tested, debugged and managed an autonomous hydroponics system.
- Arduino and Raspberry were used as microcontrollers to execute functions and remotely transfer database.

### Cadence Virtuoso Layout

Sept 2019 - Dec 2019

- Designed, simulated and prepared layout of Inverter, Flip-Flop, 2 input NAND and Mirror adder.
- Verified its efficiency and functionality using waveforms, LVS and DRC reports.

### Op-Amp Design Using PSPICE

Oct 2018 - Nov 2020

- A design-oriented trade-off was used to modify the operational amplifier to meet the desired results.
- Using a Telescopic amplifier to meet the requirement for specific voltage gain, CMRR and Unity Gain Bandwidth.
- The technology library of 0.13um for CMOS was used.

### Arduino Ultrasonic Sensor Radar

Nov 2018 - Dec 2018

- Designed Ultrasonic Sensor radar using Arduino UNO, HC-SR04 sensor, servo on breadboard with coding in C++.

### Tic Tag Toe Game (*Python*)

May 2020 - June 2020

- Developed a code for Tic Tag Toe Game that can be played by two players.
- Created 3x3 grid on which marks were made by one of two players and winner was decided on three consecutive marks.

### Simulation of Verilog design using FPGA

Jan 2018 - May 2018

- Simulated Verilog and Test bench in Modelsim to get waveform and debug circuits that are implemented by FPGA.
- Designed 8-bit Carry Adder, BCD and Arithmetic Logic design.

## PUBLICATION

Talwar R., Amala N., Medina G., Jida A.S., and Eltayeb M.E. (2020). *Exploiting Multi-Path for Safeguarding mmWave Communications Against Randomly Located Eavesdroppers*. ICSNC 2020 : The Fifteenth International Conference on Systems and Networks Communications

Link : ([https://www.thinkmind.org/articles/icsnc\\_2020\\_2\\_80\\_28008.pdf](https://www.thinkmind.org/articles/icsnc_2020_2_80_28008.pdf))